Ignorable Learning: Total Accuracy, Plan Coherence, and Conditionalization for Externalists

Abstract. If evidence can be opaque, accuracy arguments favour conditioning on the fact that one has learned something over conditioning on what one has learned. Two recent arguments defend conditionalization anyway. Meehan and Zhang rank rules by their total expected accuracy across updating problems and show that this vindicates conditionalization exactly when the weighting of problems is extensional. Zendejas Medina evaluates each response by its expected accuracy conditional on what is learned. I show that every weighting is a model of how evidence arrives and that extensionality is the statisticians' condition of coarsening at random on that model. It holds exactly when the total respects Zendejas Medina's conditional verdicts. Read as importance, the weights must satisfy it if the scores they induce are to be proper, and the total argument then shares his evaluative core. Read as frequencies, they satisfy it only on a knife-edge held by a symmetry requirement. For agents, conditionalization and its rival disagree about the world, as opposed to what one has learned, and conditionalization violates Reflection for some prior, exactly when learning is non-ignorable. Plan coherence answers the accuracy argument but leaves that cost in place.

1. Introduction

Greaves and Wallace (2006) showed that when learning is transparent, so that whenever one learns p one also learns that p is what one learned, conditionalization has the highest expected accuracy among plans that respond to what is learned. Schoenfield (2017) showed that in general the best such plan conditions instead on the proposition that one has learned p. Externalists hold that evidence is sometimes opaque, so they seem forced to give up either conditionalization or the accuracy-first view of updating, a choice Schultheis (2025) calls the Bayesian Dilemma.

Two recent arguments try to escape it. Zendejas Medina (2024) argues that plan-based arguments need a bridge from the best plan to what one does after learning, and that the arguments favouring the rival rule use the wrong bridge. The plan that binds an agent who learns p is the best plan *for* p, ranked by expected accuracy conditional on p, and that plan is to conditionalize. Meehan and Zhang (2025) instead keep unconditional expected accuracy but total it over updating problems. Among rules that respond only to the proposition learned, conditionalization maximizes the total, and it does so for a weighting of problems exactly when the weighting is *extensional* (Fact 11, 337). They leave two questions open. Is extensionality bedrock (337)? And how does their argument relate to Zendejas Medina's, given that in an opaque problem the best unconditional plan need not combine the best conditional plans (287–288 n. 2)?

I answer both by saying what extensionality is. My thesis:

> Every weighting of updating problems is a model of how evidence arrives, and extensionality is coarsening at random on that model: receiving a proposition carries no information beyond its truth. It holds exactly when the total respects Zendejas Medina's conditional verdicts. Read as importance, the weights must satisfy it if the scores they induce are to be proper,

and the total argument then shares his evaluative core. Read as frequencies, they satisfy it robustly only if no opaque problem is counted, and otherwise only through a symmetry of the standard.

A second claim generalizes the first from evaluators to agents. When opacity is ignorable, conditionalization and its rival agree about the world and differ only about what one has learned. When it is not, they disagree about the world, and conditionalization violates Reflection for some prior. So the cost of conditionalizing on opaque evidence falls where learning is non-ignorable. Zendejas Medina's plan coherence answers the accuracy argument everywhere, but it leaves this cost in place.

The statistics is not mine. Heitjan and Rubin (1991) formulated coarsening at random (CAR) as a condition for ignoring the coarsening mechanism in inference. Grünwald and Halpern (2003) used it to explain when conditioning in a space that ignores the protocol by which information arrives gives the right answer, and Gill and Grünwald (2008) characterized CAR mechanisms. Zendejas Medina already locates the threat to conditionalization in a premise of the argument rather than in opacity as such. He does not discuss learning models, totals or Reflection. What is new here is the identification of Meehan and Zhang's weightings with coarsening mechanisms; the equivalence of extensionality with respect for conditional verdicts, which links the two rebuildings; the knife-edge and neutrality results; the treatment of proposition-attached importance; and the characterization of when conditionalization satisfies Reflection.

For readers outside formal epistemology the point is this. To defend a rule "in general" one must say how much each situation counts, and any such choice behaves like a hypothesis about how one comes to learn things. The defence works exactly when that hypothesis makes the manner of learning uninformative. When it does, judging a rule as a whole agrees with judging each of its responses on its own terms.

2. Two rebuildings

I follow Meehan and Zhang's framework. A *prior space* (S, π) is a finite set of states, $\mid S \mid = n \geq 2$, with a regular probability π (290). A *learning situation* E maps each state s to the strongest proposition learned at s, a set containing s, so evidence is factive (290 n. 4). (S, π, E) is an *updating problem*. It is *transparent* if $E(s) \subseteq [E = E(s)]$ for every s, where $[E = E(s)]$ is the set of states at which $E(s)$ is learned; otherwise it is *opaque* (292–294). For factive E, transparency is partitionality (294 n. 5). Accuracy is measured by a strictly proper score A (296). BCOND recommends $\pi(\cdot \mid E(s))$, META-COND recommends $\pi(\cdot \mid E = E(s))$, and the dogmatic rule recommends π (291–292). In *Dana*, a problem about the weight of butter with prior $(0.2, 0.6, 0.2)$ over 15, 16 and 17 oz., one learns $\{15, 16\}$ at 15, everything at 16 and $\{16, 17\}$ at 17. There the dogmatic rule beats BCOND under the Brier score (Fact 3, 302).

Global evidential constancy requires that two problems on the same prior space that deliver the same proposition at a state get the same posterior there (309, 314–315). A rule satisfies it iff it is given by a map from a proposition to a posterior (Fact 5, 316); I write $f(p)$. BCOND and the dogmatic rule qualify; META-COND does not (316). *Hi-TEA* says that a rule is rational iff, among globally evidentially constant rules, it has the highest total expected accuracy relative

to every prior space (330). A *weighting* λ assigns each factive learning situation a non-negative weight, positive somewhere, and $\mathrm{TEA}_\lambda(f) = \sum_E \lambda(E) \sum_s \pi(s) A(f_{(S,\pi,E)}(s), s)$ (335). Each proposition p induces $\lambda_p(s) = \sum_{E:\, E(s)=p} \lambda(E)$, and λ is *extensional* if $\lambda_p(s)$ depends only on whether p is true at s (336); by factivity, if λ_p is constant on p. For globally evidentially constant f,

$$\mathrm{TEA}_\lambda(f) = \sum_s \pi(s) \sum_{p \ni s} \lambda_p(s)\, A(f(p), s) \qquad (*)$$

(345). By Lemma 1 (344), the optimal such rule recommends $\pi(\cdot \mid \lambda_p)$, where $\pi(s \mid \lambda_p) \propto \pi(s)\lambda_p(s)$. By Lemma 2 (345), this equals $\pi(\cdot \mid p)$ for all π and p iff λ is extensional. Hence Fact 11. Under the uniform weighting BCOND uniquely wins (Fact 10, 330). By Fact 12 (338), a point mass on one problem is extensional iff the problem is transparent.

Zendejas Medina's framework is close. His *auto-epistemic admissibility* binds an agent who learns p to the best plan conditional on her having learned p; *epistemic admissibility* binds her to the best plan conditional on p (§4.2). The two coincide when evidence is transparent. The arguments for the rival rule assume the first (§5). He argues that externalists should accept the second, since an agent who learns p without learning that she learned it is not in a position to use the latter fact (§6.1). From minimal principles about preferences between plans he derives that plans with condition p are ranked by their expected value conditional on p (§7.2). Write $\mathrm{CEA}_p(c) = \sum_{s \in p} \pi(s \mid p) A(c, s)$. By strict propriety, conditionalizing on p is then the best plan for p (§8.1). His arguments, he says, answer the question "What is the optimal way to react to the fact that E is true?" (§8.3). Meehan and Zhang describe this as a justification "based on conditional expected accuracy rather than unconditional expected accuracy", and add that "it remains a puzzle why Bayesian conditionalization gives the correct prescription for which unconditional plan a rational agent should adopt in nontransparent updating problems" (287–288 n. 2).

3. Weightings are models of learning

Observation. For every weighting and every state s, $\sum_{p \ni s} \lambda_p(s) = \Lambda$, the total weight, because each learning situation delivers exactly one proposition at s.

So $q_\lambda(p \mid s) = \lambda_p(s)/\Lambda$ is a probability distribution over the propositions true at s: a *coarsening mechanism* in the statisticians' sense, which replaces each state with a set containing it (Gill and Grünwald 2008, §2).

Proposition 1. (a) Every coarsening mechanism is q_λ for some weighting λ. (b) $\mathrm{TEA}_\lambda(f)/\Lambda$ is the expected accuracy of responding to each received proposition p with $f(p)$, for an agent with prior π who receives p at s with probability $q_\lambda(p \mid s)$. (c) The rule of Lemma 1 conditions on the event that p was received.

Proof. (a) Put $\lambda(E) = \prod_s q(E(s) \mid s)$, the distribution of a learning situation that draws a proposition independently at each state; summing over the other states' draws gives $\lambda_p(s) = q(p \mid s)$. (b) Divide $(*)$ by Λ. (c) $\Pr(s \mid p \text{ received}) \propto \pi(s) q_\lambda(p \mid s) \propto \pi(s)\lambda_p(s)$. ∎

Lemma 1 is thus the Greaves–Wallace argument run in an enlarged space whose points pair a state with a report, where the agent learns transparently which report she received. Hi-TEA selects the

rule that META-COND would recommend to an agent whose opinions about how she learns are given by q_λ. The rule is globally evidentially constant only because those opinions are fixed in the standard instead of being consulted by the rule.

A mechanism is CAR if, for each p, the probability of receiving p is the same at every state in p (Gill and Grünwald 2008, eq. 2.1). By the Observation:

> λ is extensional if and only if q_λ is CAR.

Meehan and Zhang's results then fall into place. Lemma 2 says that conditioning on the receipt of p agrees with conditioning on p, for every prior, exactly under CAR. This is the finite, Bayesian core of two classical points. Heitjan and Rubin (1991, abstract) give conditions, CAR among them, under which the coarsening mechanism "can be ignored when drawing Bayesian and likelihood inferences". Grünwald and Halpern (2003, abstract) present CAR as the condition that "characterizes when 'naive' conditioning in a naive space works". Fact 12 says that a deterministic mechanism is CAR iff it is a partition. Extensionality is therefore *transparency in distribution*. Under a CAR mechanism, learning p and learning that one has learned p need not coincide, but they have the same import for the states: receiving p is evidence of nothing beyond p.

Meehan and Zhang's own standard is a model of transparent learning. For $\varnothing \neq q \neq S$, the *yes/no problem* E_q delivers q on q and $S \setminus q$ off it, and E_S delivers S everywhere; these 2^{n-1} problems form $\mathcal{Y}$.

Proposition 2. For every globally evidentially constant f, the uniform total over all factive learning situations equals $2^{(n-1)^2}$ times the uniform total over $\mathcal{Y}$.

Proof. The uniform profile over all situations is $(2^{n-1})^{n-1}$ on p, Meehan and Zhang's α_S (344). Over $\mathcal{Y}$ it is 1 on p, since only E_p delivers p at a p-state. Apply $(*)$. ∎

The uniform mechanism colours each state red or blue with a fair coin and reports the colour of the actual state. Fact 10 then follows from Greaves and Wallace's theorem summand by summand. The result lies close to the surface of Meehan and Zhang's proof (332). It shows that their headline standard cannot tell the externalist's world from a world of random yes/no questions.

Extensional standards need not count transparent problems at all. The simplest case is a standard example in the CAR literature (Gill and Grünwald 2008, Example 1, which they credit to earlier work of Gill, van der Laan and Robins).

> *Cyclic Informants.* There are three states. Informant A reports "1 or 2" at 1, "2 or 3" at 2 and "3 or 1" at 3. Informant B reports "1 or 3" at 1, "1 or 2" at 2 and "2 or 3" at 3. Each has weight 1.

Both problems are opaque, yet each report has weight 1 at both of its states, so the weighting is CAR. Its support contains no partition. So Meehan and Zhang are right that one need not be an internalist to accept extensionality (338). What extensionality requires is that the standard's model of learning share the evidential signature of transparency.

4. Totals and conditional verdicts

Say that λ *respects conditional verdicts* if, whenever $\mathrm{CEA}_p(f(p)) \geq \mathrm{CEA}_p(g(p))$ for every p, $\mathrm{TEA}_\lambda(f) \geq \mathrm{TEA}_\lambda(g)$. A rule that does at least as well as another by Zendejas Medina's measure on every proposition is then ranked at least as high.

Proposition 3. Let π be regular. Then (i) λ is extensional iff (iii) λ respects conditional verdicts. If every λ_p is non-zero somewhere on p, both are equivalent to (ii): each induced score $A^\lambda_p(c,s) = \lambda_p(s)A(c,s)$ is strictly proper on p.

Proof. (i)⟹(iii): if $\lambda_p = k_p$ on p, $(*)$ gives $\mathrm{TEA}_\lambda(f) = \sum_p k_p \pi(p)\,\mathrm{CEA}_p(f(p))$. (iii)⟹(i): suppose λ_p is not constant on p. By regularity $c^* = \pi(\cdot \mid \lambda_p)$ differs from $c^\circ = \pi(\cdot \mid p)$. Let f be BCOND and let g differ from it only in $g(p) = c^*$. Strict propriety gives $\mathrm{CEA}_p(c^\circ) > \mathrm{CEA}_p(c^*)$. Yet $\mathrm{TEA}_\lambda(g) - \mathrm{TEA}_\lambda(f) = Z\,(\mathbb{E}_{c^*}A(c^*,\cdot) - \mathbb{E}_{c^*}A(c^\circ,\cdot)) > 0$, with $Z = \sum_{s\in p}\pi(s)\lambda_p(s)$. (i)⟺(ii): the expectation of $A^\lambda_p(c,\cdot)$ under a probability b on p is proportional to that of $A(c,\cdot)$ under b reweighted by λ_p. By strict propriety its unique maximizer is that reweighting, which is b for every b iff λ_p is constant on p. ∎

Proposition 3 resolves Meehan and Zhang's puzzle in their own terms. A single problem is a point-mass weighting, extensional only if transparent, so in an opaque problem unconditional expected accuracy must sometimes overrule a conditional verdict. A total can make the two agree, and it does so exactly when its model of learning is ignorable. The total argument vindicates the conjunction of the best conditional plans by evaluating with a model of learning under which unconditional and conditional evaluation coincide.

Proposition 3 also bears on the reading Meehan and Zhang give the weights: "the epistemic importance of updating accurately in response to learning proposition p at state s" (336). Read so, a non-extensional standard scores responses to some proposition by an improper measure, which leads an agent with any credences over the p-states to expect other credences to score better. Meehan and Zhang require strict propriety of A (296). The induced score is the measure by which the standard actually evaluates responses to p, so anyone who accepts their reading and that requirement should require it of the induced scores too, and so must accept extensionality. But then, by (iii), the total is a positive combination of conditional expected accuracies. On the importance reading, then, the total argument and Zendejas Medina's share their evaluative core. They differ in the bridge. Global evidential constancy restricts which rules are compared, where epistemic admissibility restricts which plan binds the agent.

This answers the bedrock question in one way. A view on which a rule is to be judged by how good its conditional plans are, as Zendejas Medina's framework suggests, entails (iii) and so extensionality. That grounds extensionality, but at a price for Meehan and Zhang's project: the total then adds no evaluative content to the conditional argument.

5. Importance attached to propositions

Meehan and Zhang take seriously that problems differ in importance: "[P]erhaps the updating problem where one learns the truth of some laws of nature is more important than one where one

learns the number of stones in Wisconsin" (335). Which importance weightings pass Proposition 3?

Suppose importance attaches to propositions, in the sense that the profile's dependence on the state is common to all propositions: $\lambda_p(s) = a(p)g(s)$ or $\lambda_p(s) = a(p) + g(s)$ on p, with $a(S) > 0$ and $g > 0$ in the first case. Let $\bar{a}(s) = \sum_{p \ni s} a(p)$. A rule satisfies *No News* if, on learning S, it recommends the prior.

Proposition 4. (a) For any weighting with λ_S not identically zero, the optimal rule satisfies No News iff λ_S is constant. (b) For separable profiles the budget identity fixes g: $g = \Lambda/\bar{a}$ in the multiplicative case and $g = (\Lambda - \bar{a})/2^{n-1}$ in the additive case. So extensionality, No News and constancy of $\bar{a}$ are equivalent.

Proof. (a) Lemma 1 and regularity. (b) Sum the profile over $p \ni s$ and use the Observation. Extensionality makes λ_S constant, hence g and $\bar{a}$; conversely constant $\bar{a}$ makes g, and hence every λ_p on p, constant. ∎

Natural aggregations of an importance function h on propositions have separable profiles: weight each problem by the product of the importances of what it delivers at each state, by their sum, or by the sum over the distinct propositions it delivers (the last checked by exact enumeration for $n = 3, 4$). For the product form, $\lambda_p(s) = \Lambda h(p)/H(s)$ with $H(s) = \sum_{p \ni s} h(p)$, and the optimal rule conditionalizes the *tilted prior* $\pi(s)/H(s)$. Each state has a fixed budget, since each situation delivers one proposition per state; where more important truths compete for it, each proposition's share shrinks.

> *Enough Butter.* Take Dana's prior. It matters especially to learn that there are at least 16 oz., enough for a recipe: $h(\{16, 17\}) = 5$, and $h = 1$ otherwise.

Then $H = (4, 8, 8)$ and the tilted prior is $(1/3, 1/2, 1/6)$. On learning $\{15, 16\}$ the standard favours $(2/5, 3/5, 0)$ over BCOND's $(1/4, 3/4, 0)$. On learning nothing new it moves to the tilted prior, and being globally evidentially constant it must do so in Noah's problem too, where nothing is ever learned (312). There Noah expects it to score $-53/45 \approx -1.18$ under the Brier score, against -1.12 for keeping his prior. Learning nothing is relatively more frequent at 15, where nothing important is there to be learned, so the standard treats learning nothing as evidence for 15.

Balance is easy to state. If $h(p) = h(S \setminus p)$ for every proper p, then H is constant, since each state lies in exactly one of p and $S \setminus p$. Importance that attaches to *questions* is harmless; importance that favours one answer over the other is not. The laws-versus-stones example reads naturally as a claim about questions.

6. The frequency reading: a knife-edge

Meehan and Zhang's non-Bayesian replies that "it also matters how likely it is for one to learn p at s, and perhaps also how epistemically important p is as a proposition at s" (337–338). By the Observation these are one parameter: every profile is a model of how likely one is to learn p at

s. On this frequency reading propriety has no grip, since a non-CAR profile is not an improper score but a hypothesis about how one learns, and the selected rule is the one appropriate to it. What, then, secures extensionality?

Proposition 5. Let Σ be a set of learning situations. Every weighting supported on Σ is extensional iff every member of Σ is transparent. Moreover, if λ is extensional and E is opaque, then adding weight $\varepsilon > 0$ to E yields a non-extensional weighting.

Proof. Profiles are linear in the weighting. If λ and $\lambda + \varepsilon\delta_E$ were both extensional, so would be ε times the point mass on E, contradicting Fact 12. The "if" direction is Meehan and Zhang's (336). For "only if", the point mass on any opaque member of Σ is supported on Σ. ∎

Standards that count only transparent problems vindicate BCOND however the problems are weighted. Standards that count opaque problems do so only when the opaque problems' biases cancel exactly. Add any weight to *Dana* within the uniform total and the optimal response to $\{15, 16\}$ is no longer $\pi(\cdot \mid \{15, 16\})$. The CAR literature notes the same fragility. Of *Cyclic Informants*, implemented by tossing a coin to choose the informant, Gill and Grünwald write: "If the coin is biased the CAR property can be lost" (2008, Example 2). Grünwald and Halpern (2003, abstract) find that CAR "holds rather infrequently".

What holds the edge in Meehan and Zhang's case is symmetry. A permutation σ of S acts on learning situations by $(\sigma E)(\sigma s) = \sigma[E(s)]$. Call λ *neutral* if $\lambda(\sigma E) = \lambda(E)$ for all σ and E.

Proposition 6. Every neutral weighting is extensional.

Proof. Neutrality gives $\lambda_{\sigma[p]}(\sigma s) = \lambda_p(s)$, and for $s, t \in p$ the transposition of s and t maps p to itself. ∎

The uniform total, *Cyclic Informants* and the uniform weighting on opaque problems alone are all neutral, and averaging any weighting over relabellings of the states makes it neutral. Neutrality is sufficient, not necessary: a point mass on one partition is extensional but not neutral. It supplies a criterion that does not mention conditionalization: a standard that treats a state only as the bearer of the propositions true at it vindicates BCOND.

Neutrality has a cost, and the cost locates the dispute. States are rarely bare labels. In *Dana* they are weights on a scale, and the opacity of her problem comes from that structure: one learns less near the middle of the range than at its ends. A neutral standard gives the problem in which 16 oz. is the uninformative state the same weight as the one in which 15 oz. is, so it treats the scale as if it had no shape. That is what makes neutral standards CAR. It is also what an externalist who locates opacity in margins for error has reason to doubt, if the standard is meant to reflect the situations agents are in. Meehan and Zhang suggest that Bayesianism may be a theory of "high-level" rationality coexisting with a "low-level" theory that tracks individual dispositions and environments (338). Proposition 5 gives this exact content: the high-level standard is a CAR standard, and it remains one by abstracting from the low-level structure, not by approximating it.

7. Agents: ignorability, plan coherence and Reflection

The CAR reading applies to agents as well as evaluators. Let an agent's possibilities form a finite set W with a map σ onto S, and a regular prior μ whose marginal on S is π. At each w she receives a proposition $p \ni \sigma(w)$ and learns $\sigma^{-1}[p]$, the proposition that the state is in p. Her *mechanism* is $q(p \mid s) = \mu(p \text{ received} \mid \sigma = s)$. Her evidence concerns only the state; whether it was p that she received is a further fact about W.

Proposition 7. (a) BCOND and META-COND recommend the same credences over S on every proposition received with positive probability iff q is CAR. (b) If q is CAR, BCOND satisfies Reflection for every proposition $X \subseteq W$: her prior expectation of her posterior credence in X equals her prior credence in X. Conversely, if, for fixed q, BCOND satisfies Reflection for every proposition about S and every prior on S, then q is CAR. (c) If q is CAR and some p is received with probability $q_p < 1$ at the states in p, then on receiving p BCOND gives credence q_p to the proposition that p is what she received.

Proof. (a) META-COND's credence in $s \in p$ is proportional to $\pi(s)q(p \mid s)$, and BCOND's to $\pi(s)$. (b) Let μ_p be the probability of receiving p. The expected posterior in X is $\sum_p \mu_p\, \mu(X \cap \sigma^{-1}[p])/\pi(p)$. Under CAR $\mu_p = q_p\pi(p)$, so this is $\sum_{w \in X} \mu(w) \sum_{p \ni \sigma(w)} q_p = \mu(X)$. Conversely, Reflection for each state t says $\sum_{p \ni t} \mu_p/\pi(p) = 1$. With π replaced by an arbitrary positive vector x, each term is $\sum_{s \in p} x_s q(p \mid s) / \sum_{s \in p} x_s$, and the identity holds on an open set, hence identically. Clearing denominators, each linear form $\sum_{s \in p} x_s$ with $\mid p \mid \geq 2$ is irreducible and not proportional to the others, so it divides its own numerator. Both are linear, so the numerator is a multiple of it, and $q(p \mid \cdot)$ is constant on p. (c) $\mu(p \text{ received} \mid \sigma^{-1}[p]) = q_p\pi(p)/\pi(p)$. ∎

I checked (a), the first half of (b), and the converse's solution space for $n = 3, 4$ by computation. Part (a) is Grünwald and Halpern's characterization restated. It answers, for credences about the state, a question Das raises: under what conditions conditionalizing on a class of propositions maximizes expected accuracy (Das 2019, §5.3). Among plans that respond to what is received, META-COND's credences about S are optimal by strict propriety. By (a), BCOND's coincide with them iff learning is ignorable. I have not found (b) stated in this form.

Proposition 7 separates two components of the externalist's problem. Take *Cyclic Informants* with a fair coin. The agent's evidence is opaque: after learning that the state is in p, BCOND leaves her, by (c), at credence $1/2$ that this is her evidence. So she is rationally uncertain what her evidence is. META-COND makes her certain of it. Yet by (a) the two rules agree about the state, and by (b) she cannot expect her opinion about anything to move in any direction. Under ignorable learning the whole disagreement concerns what one has learned. Under non-ignorable learning it reaches the world. Margin-for-error cases are non-ignorable by construction: whether one learns p depends on where within p one is. So is Zendejas Medina's own example. In *Here's a Hand*, the veridical case delivers that one has a hand and the deceived case delivers only the experience (§3.2). Add that without the experience one learns that one lacks it. The mechanism is then deterministic and not a partition, so it is not CAR. With a uniform prior over the good case, the bad case and no experience, a conditionalizer's expected posterior that she has a hand is $1/2$, against a prior of $1/3$.

This is the cost that Gallow (2021, abstract) points to when he argues that "if externalism is correct, then the orthodox Bayesian learning norms of conditionalization and reflection are inconsistent with each other". Proposition 7(b) locates the conflict: the two norms clash for some prior exactly when learning is non-ignorable. It also bears on Salow's (2018) argument that "intentionally biased inquiry is impossible", which he takes to support an access principle. An agent whose learning is opaque but ignorable violates access, yet can never expect conditionalization to move her credence in a given proposition on balance. Whether that suffices to block his argument depends on details I do not examine. Gallow's own remedy replaces conditionalization; Zendejas Medina notes that it collapses into the auto-epistemic rule for ideal agents who know they will follow their plan (§8.4).

The two diagnoses can now be compared exactly. Zendejas Medina locates the threat in a bridge premise, auto-epistemic admissibility, and his remedy works for every mechanism. It removes the accuracy argument's force against conditionalization, including where the rules disagree about the world. It does not restore Reflection, which he does not discuss. Ignorability locates the threat in the learning model. Where learning is ignorable, both troubles about the world, the disagreement and the failure of Reflection, vanish without any premise about plans, but the remaining disagreement about one's evidence needs his kind of answer. The externalist's paradigm cases (margins for error, sceptical scenarios) are non-ignorable. So an externalist who adopts his remedy keeps conditionalization at the price of Reflection in exactly those cases. Meehan and Zhang report that Carr's (2021) formally similar aggregate represents "the agent's subjective estimate of how accurate an updating rule is, given her uncertainty about what she might learn at a given state" (330 n. 35). On that reading, Proposition 7(a) applies to her aggregate directly.

8. Objections and replies

Objection 1: plan coherence makes ignorability idle. If Zendejas Medina is right, the agent who learns p is bound by the best plan for p, whatever the mechanism. The divergence from META-COND then reflects only a mistaken bridge principle. So CAR marks nothing of normative significance, and Meehan and Zhang's total is beside the point.

Reply. The objection succeeds against one claim and fails against three. It succeeds against the claim that a vindication of conditionalization requires ignorable learning; it does not, if epistemic admissibility is granted. First, it does not touch the analysis of Meehan and Zhang. Their argument was meant to work through unconditional accuracy without such a premise. Proposition 3 shows that it reaches conditionalization only by a total under which unconditional evaluation reproduces his. Second, plan coherence says which plan binds the agent. It says nothing about what her plan-guided credences expect of themselves, and by Proposition 7(b) the failure of Reflection is fixed by the mechanism. Third, his choice of condition is the agent-level counterpart of the CAR standard. Ranking responses by CEA_p is what an agent would do who thought that how p arrives carries no information. His argument for that choice, that one has not learned that one learned p, is stronger than a symmetry requirement on populations. But it is an argument that the manner of learning may not be used, not that it is uninformative. Where it is informative, an agent with a suitable prior can foresee the direction in which conditionalizing will move her.

Objection 2: the knife-edge is harmless. The optimal rule $\pi(\cdot \mid \lambda_p)$ varies continuously with λ. With weight $1/10$ added to *Dana* in the uniform total, the optimal response to $\{15, 16\}$ is $(161/641, 480/641, 0)$, against $(1/4, 3/4, 0)$.

Reply. Granted, as far as it goes: near-CAR standards still defeat the truth rule. But Hi-TEA claims that BCOND is uniquely rational, and that claim survives only at the edge. Departures that reflect where agents are need not be small. If *Dana* gets as much weight as the other 63 problems together, the optimal response to $\{15, 16\}$ becomes $(5/8, 3/8, 0)$, and on learning nothing the rule moves from $(0.2, 0.6, 0.2)$ to $(1/17, 15/17, 1/17)$. An approximate vindication also needs a reason for the standard to lie near a CAR one, and that reason is again neutrality.

Objection 3: why sum at all? A cautious evaluator might rank rules by their worst case.

Reply. The worst expected accuracy of f over factive situations is $\sum_s \pi(s) \min_{p \ni s} A(f(p), s)$, since a situation chooses the proposition at each state independently. By propriety this is at most the dogmatic rule's value. With Dana's prior and the Brier score, BCOND's worst case is -1.188, in Dana's problem, against -1.12. Minimax regret fares no better: since the best globally evidentially constant rule in any problem matches META-COND there (327), BCOND's maximal regret is 1.188 and the dogmatic rule's 1.12. Dominance does not single out BCOND either; it is undominated (Fact 7, 324), but so is the one-shot rule tailored to Dana (326 n. 32). A vindication along these lines must aggregate by weights.

9. Conclusion

Meehan and Zhang showed that their argument stands or falls with extensionality. I have argued that extensionality is coarsening at random for the model of learning that every weighting implicitly is. It holds exactly when the total respects Zendejas Medina's conditional verdicts, so where it holds the two rebuildings evaluate the same thing. Read as importance, the weights must be extensional if their induced scores are to be proper, and the total then adds no evaluative content to the conditional argument. Read as frequencies, they are extensional robustly only if no opaque problem is counted, and otherwise through the symmetry of the standard. Proposition-attached importance preserves extensionality exactly when No News holds. For agents, conditionalization and its rival disagree about the world, and conditionalization violates Reflection for some prior, exactly when learning is non-ignorable. Plan coherence answers the accuracy argument but leaves that cost where it was.

I have not shown that neutrality is mandatory, nor that Reflection should outrank plan coherence. My results are for finite spaces and factive evidence; for infinite non-partitional settings, see Easwaran and Nielsen (2025).

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